

Test Project: Instagram Clone

Objective

Build a simplified Instagram-like web application that includes core social media features such as user accounts, posts, interactions (likes, comments, replies), stories, scheduling, and notifications. This project will evaluate your understanding of full-stack development, architecture, code quality, and your ability to implement a minimal user interface in React.

Core Features & Requirements

1. Authentication

- Users should be able to register, log in, and log out.
- Support email/password-based authentication.
- Implement JWT or session-based authentication.
- Ensure secure password hashing (e.g., bcrypt).

2. Authorization

- Only authenticated users should be able to access the platform.
- Users must be able to create, edit, and delete their own posts, comments, and stories.
- Unauthorized actions should return appropriate error messages.
- Users should only be able to view posts of people they are connected with.

3. Post Creation

- Users can create image-based posts.
- Each post should have the following:
 - A caption
 - A timestamp
 - An image upload feature (either through cloud storage or local storage for simplicity).

- Posts should be visible in reverse chronological order (latest first).

4. Connections (Follow/Unfollow)

- Users should be able to follow and unfollow other users.
- Display follower and following counts on user profiles.
- A user's feed should display posts only from people they follow.

5. Likes

- Users can like and unlike posts.
- The like count should be visible on each post.
- Users should be able to view who liked a particular post.

6. Comments & Replies

- Users can comment on posts.
- Users can reply to other comments (nested replies allowed, but limited to one level).
- Comments should be displayed under their respective posts.
- Users should only be able to delete their own comments or replies.

7. Stories (Auto-delete after 24 hours)

- Users can upload a story, which may be an image or a short video.
- Stories should be visible for 24 hours and automatically deleted thereafter.
- Users should be able to see stories of people they follow in a "story bar."

8. Scheduled Posts

- Users should be able to schedule posts for future publishing.
- Provide a user interface to set the date and time for scheduled posts.
- Scheduled posts should appear in the feed only after the specified scheduled time.

9. Notifications

- Send notifications for:
 - New followers
 - Other relevant activities such as comments and likes.
 - Notifications should be displayed in a dropdown or dedicated page.
 - Provide an option to mark notifications as read.
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Additional Requirement: Minimal UI in React

- Implement a minimal front-end UI using React.
- The UI should allow users to interact with the core features of the application: logging in, creating posts, following/unfollowing users, liking posts, commenting, and viewing stories.
- Ensure the UI is responsive and user-friendly, providing an intuitive experience for users.